Example 5.7

Show that the points $A(-4,-3)$, $B(3,1)$, $C(3,6)$, $D(-4,2)$ taken in that order form the vertices of a parallelogram.

Solution

Let $A(-4, -3)$, $B(3, 1)$, $C(3, 6)$, $D(-4, 2)$ be the four vertices of any quadrilateral $ABCD$. Using the distance formula,

Parallelogram

We know that opposite sides are equal

Let $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$AB = \sqrt{(3+4)^2 + (1+3)^2} = \sqrt{49 + 16} = \sqrt{65}$

$BC = \sqrt{(3-3)^2 + (6-1)^2}$

$= \sqrt{0 + 25} = \sqrt{25} = 5$

$CD = \sqrt{(-4-3)^2 + (2-6)^2}$

$= \sqrt{(-7)^2 + (-4)^2} = \sqrt{49 + 16} = \sqrt{65}$

$AD = \sqrt{(-4+4)^2 + (2+3)^2} = \sqrt{(0)^2 + (5)^2} = \sqrt{25} = 5$

$AB = CD = \sqrt{65}$ and $BC = AD = 5$

Here, the opposite sides are equal. Hence $ABCD$ is a parallelogram.

Example 5.8

Calculate the distance between the points A (7, 3) and B which lies on the x-axis whose abscissa is 11.

Solution

Since B is on the x-axis, the y-coordinate of B is 0.

So, the coordinates of the point B is (11, 0)

By the distance formula the distance between the points A (7, 3), B (11, 0) is

$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$AB = \sqrt{(11-7)^2 + (0-3)^2} = \sqrt{(4)^2 + (-3)^2} = \sqrt{16+9} = \sqrt{25}$

$= 5$

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